Machine Learning Coursework 2

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1 Introduction

Deep learning systems depend on large amounts of annotated data. However, acquiring labels is expensive and time-consuming, particularly in specialist domains. Hacohen et al. [1] address this by introducing TypiClust, an active learning strategy for low budgets that selects a small, diverse set of typical examples representative of the broader data distribution. This reduces annotation cost while maintaining classifier performance.

2 Methodology

Active learning allows a model to select which data points are labelled under a fixed budget. Traditional methods use uncertainty sampling, but in the low-budget regime neural networks produce unreliable confidence estimates, this is known as the cold start problem. To overcome this, TypiClust selects data points based on typicality, defined as the density of a point within a learned feature space [1]. For low budgets it is best to sample the most typical data points.

The TPC_{RP} variant of TypiClust consists of three main steps. First, a ResNet encoder is trained on the full unlabelled dataset using SimCLR, and embeddings are extracted from the penultimate layer. Second, for each of the B data points to be selected in an active learning iteration, K-Means re-clusters the embeddings into $|L| + B$ clusters, where L is the set of labelled examples and B is the budget. Third, the algorithm identifies the largest cluster with the fewest labelled points (minimum size > 5) and selects the most typical unlabelled point from it. That point is added to L and removed from U, the set of unlabelled examples. This process is repeated for a fixed number of active learning iterations.

The full implementation of TPC_{RP} is available at: `https://github.com/katiepreed/COURSEWORKML2`.

3 Results

TypiClust was evaluated on CIFAR-10 across three frameworks: fully supervised, fully supervised with self-supervised embeddings, and semi-supervised using FlexMatch (Figure 1). The experimental setup follows the original paper, which uses external codebases from Munjal et al. [2], Van Gansbeke et al. [3], and Zhang et al. [4]. All code was independently re-implemented to develop a deeper understanding of the underlying methods.

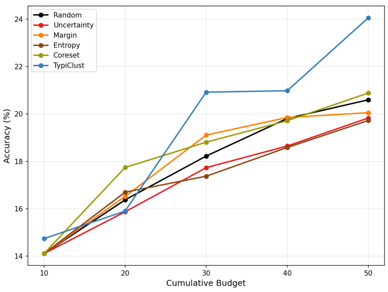
Due to computational constraints, a subset of the baselines used in the original paper were selected for evaluation. The chosen strategies are sufficient to demonstrate the effectiveness of TypiClust. Non-TypiClust baselines default to random selection in the first iteration, as shown in Figures 1a and 1b.

Strategy	**Fully Supervised**	**Self-Supervised Embeddings**	**Semi-supervised**
Random	20.60 ± 1.58	75.51 ± 2.24	10.00
Uncertainty	19.83 ± 1.82	70.62 ± 5.94	N/A
Margin	20.05 ± 1.96	77.76 ± 3.11	N/A
Entropy	19.73 ± 2.00	67.55 ± 3.54	N/A
Coreset	20.88 ± 1.50	75.46 ± 1.92	10.00
Balanced	N/A	N/A	10.00
TypiClust	$\mathbf{24.06 \pm 0.00}$	$\mathbf{83.23 \pm 0.00}$	**12.67**

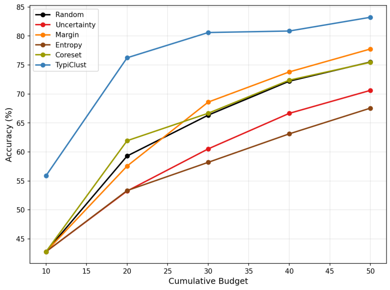
Table 1: Top-1 Test Accuracy (%) at final budget

Uncertainty-based methods perform at or below the random baseline, while coreset and margin show only marginal improvements, confirming the cold-start problem as described in the paper. This shows that conventional active learning strategies are counterproductive. TypiClust consistently outperforms random selection across all three frameworks, as shown in Figure 2a. The advantage is largest in the self-supervised embeddings setting at budget 10, where it outperforms the random baseline by 13.11%, as shown in Table 2. This is consistent with the paper's claim that TypiClust is most useful in low-budget settings.

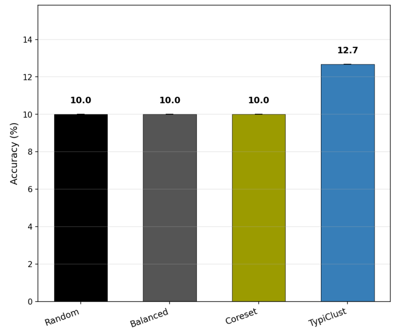
TypiClust was evaluated with a fixed clustering seed to ensure reproducibility. As a result, the selection pipeline is fully deterministic, which is why it has a standard deviation of zero. Uncertainty based methods show high variance (e.g. uncertainty: 70.62 ± 5.94 in embeddings setting), demonstrating that they are unstable in low-budget regimes. In figure 1a, TypiClust briefly underperforms at budget 20. This performance drop reflects the fixed properties of those specific points due to TypiClust's deterministic selection, and is resolved at higher budgets.



(a) Fully supervised framework



(b) Fully supervised with self-supervised embedding framework



(c) Semi-supervised framework

Figure 1: Accuracy vs. cumulative budget across three evaluation frameworks on CIFAR-10.

The fully supervised and self-supervised embeddings results closely match those reported in the original paper. However, due to computational constraints, the semi-supervised experiment was run for only 100k iterations compared to the paper's 400k, and with a single repetition rather than three, resulting in lower accuracy across all strategies in this framework.

4 Algorithm Modification

TypiClust can be improved by replacing K-Means clustering and KNN-based typicality with a Gaussian Mixture Model (GMM), while still preserving the original cluster selection logic. Cluster assignments are derived from GMM components, and typicality is measured by log-likelihood. Because the 512-dimensional SimCLR embedding space is too large for stable covariance estimation, random projection to 50 dimensions is applied prior to fitting the GMM, which was found to have the best overall performance among different reduction methods, Figure 2b.

K-Means assumes spherical, equally sized clusters [5], which is an assumption unlikely to hold in a SimCLR feature space. GMM handles elliptical clusters of different sizes [6], and its log-likelihood provides unified density estimate, eliminating the need for a separate KNN typicality computation. A high log-likelihood indicates that a point lies in a dense region of the overall data distribution

Both variants were evaluated against the self-supervised embeddings experiment under identical conditions: 5 active learning iterations, with a budget of 10 per iteration, for 400 epochs. The modification improved performance by 3.91% on average. At budget 10 the difference in accuracy is the largest, with TypiClustGMM achieving 16.35% increase in accuracy, as shown in figure 2c. This improvement at the lowest budget is particularly significant, as this is where sample selection matters most.

A limitation is that GMM log-likelihood is a global density estimate, meaning it evaluates a data point against all Gaussian components in the mixture. So a point near the boundary of two components might receive a larger log-likelihood score because both contribute. With few clusters, GMM log-likelihood helps identify representative points but with higher cluster counts, boundary effects cause performance to converge with K-Means. Future work could explore component-conditional likelihood, where the density of a point is computed using only the Gaussian component it is assigned to.

5 Conclusion

This report demonstrates that TypiClust consistently outperforms uncertainty-based and random active learning strategies in the low-budget regime on CIFAR-10, demonstrating its potential for reducing labelling costs in specialist domains. To further improve the algorithm, avenues of research include self-supervised methods that outperform SimCLR (e.g. VICReg [7]) and alternative backbone architectures for richer feature extraction.

References

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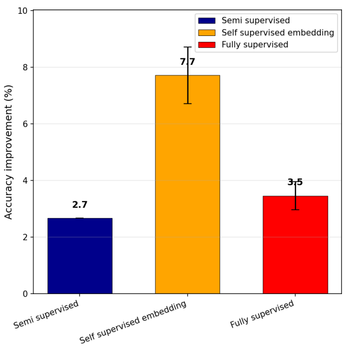
[7] Uelwer, T., Robine, J., Wagner, S. S., Höftmann, M., Upschulte, E., Konietzny, S., Behrendt, M., & Harmeling, S. (2025). A survey on self-supervised methods for visual representation learning. Machine Learning, 114, Article 111. https://doi.org/10.1007/s10994-024-06708-7

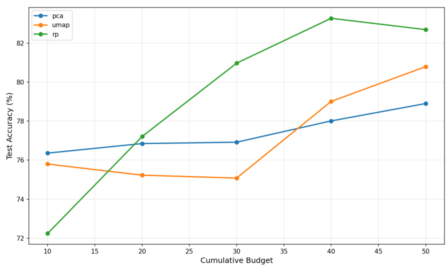
A Appendix

Strategy	**10**	**20**	**30**	**40**	**50**
Random	42.78 ± 5.94	59.29 ± 4.58	66.34 ± 4.60	72.20 ± 2.84	75.51 ± 2.24
Uncertainty	42.78 ± 5.94	53.25 ± 6.88	60.52 ± 6.11	66.65 ± 7.77	70.62 ± 5.94
Margin	42.78 ± 5.94	57.56 ± 6.20	68.59 ± 5.21	73.80 ± 5.08	77.76 ± 3.11
Entropy	42.78 ± 5.94	53.33 ± 6.73	58.20 ± 5.31	63.12 ± 4.74	67.55 ± 3.54
Coreset	42.78 ± 5.94	61.93 ± 2.42	66.67 ± 3.53	72.38 ± 2.50	75.46 ± 1.92
TypiClust	$\mathbf{55.89 \pm 0.00}$	$\mathbf{76.24 \pm 0.00}$	$\mathbf{80.60 \pm 0.00}$	$\mathbf{80.86 \pm 0.00}$	$\mathbf{83.23 \pm 0.00}$

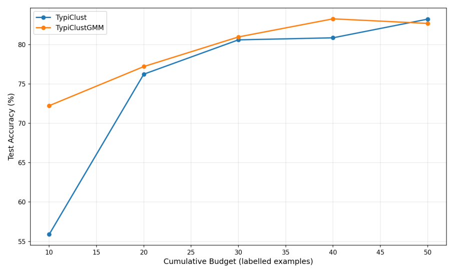
Table 2: Fully supervised with self-supervised embedding (Top-1 Test Accuracy at cumulative budget)



(a) TypiClust vs random selection baseline accuracy improvement for final budget



(b) Dimensionality reduction comparison



(c) TypiClust vs TypiClustGMM

Figure 2: Additional plots